

Inverse of Matrices

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Learning Outcomes

At the end of this lecture, you should be able to:

- ☐ Know what is the inverse of a matrix and know that not all matrices are invertible.
- ☐ Understand and the properties of invertible matrices.
- ☐ Know how to find the inverse of a 2×2 matrix.
- ☐ Know how to find the power of a matrix.

Introduction

Consider the equation

$$ax = b.$$

for real numbers a, b, x with $a \neq 0$. How do we solve this for x ?

We multiply both sides of the equation by a^{-1} and we get

$$a^{-1}ax = a^{-1}b.$$

Then we simplify $a^{-1}a = 1$ and we have

$$x = a^{-1}b \text{ or } x = \frac{b}{a}.$$

Could we do the same for matrices?

Invertible Matrix

An n -square matrix $A \in M_n(\mathbb{F})$ is said to be **invertible** if there exists a n -square matrix $B \in M_n(\mathbb{F})$ such that $AB = I_n$ and $BA = I_n$. Such a matrix B is called an **inverse** of A .

If a matrix A has an inverse, then A is also called a **nonsingular matrix**.
If A has no inverse, then A is called a **singular matrix**.

Example: Let $A = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 5 & 3 \\ 1 & 0 & 8 \end{bmatrix}$ and $B = \begin{bmatrix} -40 & 16 & 9 \\ 13 & -5 & -3 \\ 5 & -2 & -1 \end{bmatrix}$.

Invertible matrix

Example: Consider the matrix equation $\begin{bmatrix} -40 & 16 & 9 \\ 13 & -5 & -3 \\ 5 & -2 & -1 \end{bmatrix} X = \begin{bmatrix} 4 \\ 1 \\ 0 \end{bmatrix}.$

Invertible matrix

Example: Show that $A = \begin{bmatrix} 1 & 0 \\ 1 & 0 \end{bmatrix}$ is singular.

Properties of inverse of matrix

Uniqueness of inverses:

Let $A \in M_n(\mathbb{F})$. If A is invertible, then its inverse is unique, i.e, if B and C are inverses of A , then $B = C$.

So we can use A^{-1} to denote this unique inverse of A .

Cancellation Law:

Let A be an invertible m -square matrix.

- (i) If B_1 and B_2 are $m \times n$ matrices with $AB_1 = AB_2$, then $B_1 = B_2$.
- (ii) If C_1 and C_2 are $n \times m$ matrices with $C_1A = C_2A$, then $C_1 = C_2$.

In general, the Cancellation Law may not hold.

Example: Take $A = \begin{bmatrix} 1 & 0 \\ 1 & 0 \end{bmatrix}$, $B_1 = \begin{bmatrix} 2 & 0 \\ 0 & 1 \end{bmatrix}$ and $B_2 = \begin{bmatrix} 2 & 0 \\ 1 & 3 \end{bmatrix}$. Then $AB_1 = AB_2$, but $B_1 \neq B_2$.

Properties of inverse of matrix

Let $A, B \in M_n(\mathbb{F})$ be n -square invertible matrices. Then the following properties hold.

(i) λA is invertible and $(\lambda A)^{-1} = \frac{1}{\lambda} A^{-1}$ for some nonzero scalar λ .

(ii) If A^T is invertible and $(A^T)^{-1} = (A^{-1})^T$.

(iii) A^{-1} is invertible and $(A^{-1})^{-1} = A$.

(iv) AB is invertible and $(AB)^{-1} = B^{-1}A^{-1}$.

If $A_1, A_2, \dots, A_k$ are invertible matrices of the same size, then $A_1 A_2 \cdots A_k$ is invertible and $(A_1 A_2 \cdots A_k)^{-1} = A_k^{-1} A_{k-1}^{-1} \cdots A_2^{-1} A_1^{-1}$.

Properties of inverse of matrix

Example:

Let $A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$ and $ad - bc \neq 0$. Show that A is invertible and

$$A^{-1} = \frac{1}{ad-bc} \begin{bmatrix} d & -b \\ -c & a \end{bmatrix}.$$

Properties of inverse of matrix

Example:

Let A be an n -square matrix such that $A^2 - 3A - 6I_n = 0_n$. Show that A is invertible.

Power of matrices

Let $A \in M_n(\mathbb{F})$ and let k be a non-negative integer. We define A^k as follows:

$$A^k = \begin{cases} I_n & \text{if } k = 0 \\ \underbrace{AA \dots A}_{k \text{ times}} & \text{if } k \geq 1 \end{cases} .$$

If A is invertible, then for $k \geq 1$, $A^{-k} = (A^{-1})^k = \underbrace{A^{-1}A^{-1} \dots A^{-1}}_{k \text{ times}}$

Let A be an n -square matrix and let r and s be integers. Then $A^r A^s = A^{r+s}$.

If A is invertible, then A^k is invertible and $(A^k)^{-1} = A^{-k}$, where k is a non-negative integer.

Power of matrices

Example:

$$\text{Let } A = \begin{bmatrix} 1 & 2 \\ 1 & 3 \end{bmatrix} \text{ and } B = \begin{bmatrix} 3 & -2 \\ -1 & 1 \end{bmatrix}.$$

Power of matrices

Example:

Let A and B are invertible, simplify

$$M = B^2(AB)^{-1}A(B^{-1})^T(AB)^T$$

Power of matrices

Example:

Suppose that A is invertible. Given that $A^2 = 2A + 5I$, express A^4 and A^{-1} as linear combination of A and I .